Voice Scribe — Dataset Selection & English Benchmark v1.1

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Summary

We have a fixed, reproducible English test set (v1.1) for choosing a speech-to-text model: 8.6 h of full consultations, 570 noisy and background-talker versions of them, 8.4 h of Indian clinical speech and 9.6 h of Indian-English speech. It is built on Aditi's laptop in `C:\Users\aditi\voice_scribe_data\benchmark_v1\data\` and verified against a reference build.

- **Datasets chosen:** PriMock57 (consultations), Eka Medical ASR (Indian doctors, drug names), Svarah (Indian accents), MS-SNSD (noise). DISPLACE-M (real Hindi clinical audio) is requested but gated.
- **Key experiment:** each consultation is mixed with clinic noise and another conversation; the reference stays consultation-only, so words picked up from the background count as errors.
- **Not started:** model testing, Hindi/Kannada data, own OPD recordings.

Problem and requirements

The scribe must transcribe only the doctor–patient conversation in a noisy Indian OPD, in English first, then Hindi, Kannada and code-switched speech. To compare models fairly, a test set needs:

1. Doctor–patient consultation speech with medical vocabulary.
2. Indian accents and Indian drug names.
3. Clinic noise and other people talking nearby.
4. A reference transcript of the consultation only, so leaked background speech is measurable.
5. A dev/test split with no speaker in both.

No public dataset has all five. Real ambient clinic audio is rarely released because of privacy law, so the set combines datasets that each cover part of the list, plus controlled mixing for items 3 and 4.

Dataset selection

Eight datasets made the final list, each for one part of the problem; four are built into v1.1 and the rest wait on access or later phases.

Dataset	Phase	Covers	Access	Status
PriMock57	English	57 full consultations, separate doctor/patient tracks, timestamped transcripts (British accents)	Open, CC-BY-4.0	Built
Eka Medical ASR	English	Indian doctors; drug, finding and dosage terms with entity tags	Open, MIT	Built
Svarah	English	Indian English, 117 speakers, 19 states (not medical)	Terms on HF, CC-BY-4.0	Built
MS-SNSD	English	Real noise: AC hum, babble, doors, typing, announcements	Open, MIT	Built (noise source)
DISPLACE-M 2026	Hindi	~35 h real health-worker–patient Hindi/Hinglish, far-field, noisy	Gated: mentor signs terms	To request
DISPLACE 2024	Kannada	Far-field group talk in Kannada, Hindi, Telugu, Bengali, Indian English	Password from organisers	Later
IndicVoices	Hindi, Kannada	Mostly unscripted speech, 22 languages	Terms on HF, CC-BY-4.0	Later
Own staged OPD recordings	All	Real rooms, real nearby conversations	Consent needed	Not started

Rejected:

- **Synth-DoPaCo, United-Syn-Med:** synthetic speech flatters every model.
- **AfriSpeech-200:** read speech, African accents, non-commercial licence.
- **AMI, VoxConverse:** non-medical; useful later for diarization only.

- **bala-ceg Indic dataset:** one one-minute clip per language, the same translated script.
- **MeDial-Speech:** page still says it's under preparation.
- **VAANI, MUCS 2021:** kept as options for dialect and code-switching checks, not needed yet.

Corrections to the partner's research (repo L9G9N0/voice_scribe):

- The bala-ceg dataset from the project plan was missing from the catalogue.
- DISPLACE-M is Hindi-only in Phase I; no Kannada.
- Model testing does not need to wait for gated data: PriMock57, Eka and Svarah are open.
- Eka: the report called it single-speaker dictation. I first said it was mostly conversation, from preview rows. The full metadata shows both were partly wrong: 2,206 single-term clips, 1,303 dictated sentences, 110 conversation clips (0.45 h).

The benchmark build (v1.1)

All audio is 16 kHz mono, every dataset has one manifest CSV, and about 20% of each dataset is dev (for tuning), the rest test (for reporting).

Part	Size	Dev / test	Split unit
PriMock57 consultations (doctor + patient mixed)	57, 8.6 h	11 (1.7 h) / 46 (7.0 h)	consultation, stratified by recording day
PriMock57 utterances (separate tracks)	6,551	same as consultation	consultation
PriMock57 noisy versions	570 files	same as source consultation	consultation
Eka Medical ASR (en)	3,619 clips, 8.4 h	735 / 2,884	speaker; conversation clips by session (no speaker id)
Svarah	6,656 clips, 9.6 h	1,351 / 5,305	speaker (demographic profile as proxy)

Per-dataset preparation:

- **PriMock57:** doctor and patient tracks summed into one consultation track; TextGrid transcripts merged in time order; `<UNSURE>` tags kept as words, `<UNIN/>` (unintelligible) removed. Utterances are flagged when they contain either (15% and 17%), so a strict score can exclude them. 176 filler-only utterances dropped.
- **Eka:** MP3 decoded to FLAC; speaker e-mail addresses replaced with `eka_spk_NN`; entity tags kept per clip. Dev speakers chosen so each clip type (single term, sentence, conversation) is about 20% dev.
- **Svarah:** no speaker ids are released; 115 demographic profiles (for 117 speakers) used as the split unit, so two speakers sharing a profile stay together.

Noise simulation (ClinicalMix v1.1): 10 versions of each consultation. The reference is always the original consultation.

Condition	Background conversation (dB below target)	Clinic noise (dB below target)
noise_snr20 / 10 / 5	–	20 / 10 / 5
talker_sir10 / 5 / 0	10 / 5 / 0, British (another PriMock57 consultation, same split)	–
talker_in_sir10 / 5 / 0	10 / 5 / 0, Indian (two Svarah speakers taking turns, same split)	–
opd_sir5_snr10	5, British	10

- **Noise bed:** AC hum + ambient babble or hallway + an event (door, typing, announcement, copier) every 15–45 s. Dev and test draw from different MS-SNSD clips.
- **Levels:** measured against the target's annotated speech regions. Every gain, offset, noise clip and background-conversation id is stored per file, so leakage of background words can be measured directly.

Text normalisation v1 (applied to reference and model output alike): lowercase, punctuation stripped, hyphens to spaces, fillers (um, uh, erm, hmm, mm, mhm…) removed, numbers left as written.

Verification

The laptop build matches the reference build: four manifests byte-identical, the noisy manifest identical except gains that differ by at most one part in a million.

Checks run:

- **Audio integrity:** all 114 PriMock57 WAVs checked against their header length; Eka decoded durations match the source metadata (max difference 0.001 s).
- **Mixing levels:** measured on a sample consultation. Noise conditions hit 20.0 / 10.0 / 5.0 dB exactly; British talker within 0.3 dB; Indian talker within 0.1 dB.
- **Split leakage:** no Eka or Svarah speaker, and no Eka conversation session, appears in both dev and test.
- **Reproducibility:** a second full build gave byte-identical manifests and audio. `run_all.py` re-checks every new build against `reference_md5.txt`.

Issues found and fixed:

Issue	Cause	Fix
3 consultations 41–44 s instead of 7–9 min	Six WAVs half-downloaded by an earlier interrupted attempt; the check only read the header	Check now compares file size with the header's declared size; re-downloaded and rebuilt
Noisy manifest flagged as different	Windows numpy sums float32 in a different order (~1e-6 relative, ~0.00001 dB)	Gains compared with tolerance; all other fields still exact
All Eka conversation clips in dev	Those clips have no speaker id, so they formed one group	Split by recording session instead
Svarah has no speaker ids	Not released	Demographic profile used as proxy

Folder layout and rebuild

Everything lives in `C:\Users\aditi\voice_scribe_data\`; manifest paths are relative to `benchmark_v1\data\`.

```
voice_scribe_data/
  eka_medical_asr/           source download (hf)
  svarah/                    source download (hf)
  sources/
    primock57_data/          PriMock57 transcripts + clinician notes
    ms_snsd_noise/           MS-SNSD noise_train + noise_test
  benchmark_v1/
    README.md                every column explained
    scripts/                 build scripts, normalize.py, run_all.py,
reference checks
    data/
      primock57/             audio/ mixed/ noisy/ references/ notes/ + 3
manifests
      eka_en/                audio/ + manifest.csv
      svarah/                audio/ + manifest.csv
```

To rebuild or regenerate on another machine:

```
python -m pip install numpy soundfile pyarrow pandas
python benchmark_v1\scripts\run_all.py
```

The script downloads PriMock57 audio, skips steps already done, rebuilds anything derived from changed audio, and ends with a pass/fail check. `sources\MS-SNSD` and `sources\primock57` are empty leftovers and can be deleted.

Limitations and open questions

Limitations:

- PriMock57 is British-accented, clean remote-call audio: it tests long consultations and noise robustness, not Indian accents.
- Eka is mostly narration (single terms and dictated sentences); only 0.45 h is conversation. Report WER per clip type.
- PriMock57 releases no doctor ids, and its 7 clinicians recur across dev and test. Fine for comparing pretrained models, not for training.
- Mixing is digital: no room echo yet, and nearby talkers come from recordings, not a real room.
- No Hindi, Kannada or code-switched data is in the set yet.

Open questions:

- How to score numbers and units in Eka ("500 mg" vs "five hundred milligrams", "3RD DAY").
- Where models will run: college GPU server, Colab, or cloud APIs.

Next steps

Model selection on the English set comes next; Hindi and Kannada follow once DISPLACE access arrives.

- [ ] Send the DISPLACE-M and DISPLACE 2024 registration forms, with mentor signature
- [ ] Accept IndicVoices terms on Hugging Face
- [ ] Decide compute: college GPU, Colab, or cloud APIs
- [ ] Settle number/unit normalisation for Eka
- [ ] Choose candidate models: Whisper large-v3, IndicConformer, a cloud service, Gemini, Sarvam, IndicWhisper, Eka's Parrotlet-a-en-5b
- [ ] Write the scoring script: WER/CER, medical-term accuracy, background-speech leakage, speed
- [ ] Run candidates on dev, then report test
- [ ] Plan 20–30 staged OPD recordings: consent, phone placement, script outline

Reference point: Eka's own published results on its full English set are Whisper large-v3 15.7% WER, Gemini 2.5 Flash 14.8%, AWS Transcribe 18.3%, Parrotlet-a-en-5b 10.9%. On real Indian hospital recordings, ASR Under the Stethoscope found Gemini 2.5 Pro best (14.2% English, 18.5% Hindi, 35.0% Kannada) and Whisper large-v3 near 98.6% on Kannada.

Sources and licences

Source	Licence	Citation
PriMock57	CC-BY-4.0	Papadopoulos Korfiatis et al., ACL 2022
Eka Medical ASR	MIT	Eka Care
Svarah	CC-BY-4.0	Javed et al., Interspeech 2023
MS-SNSD	MIT	Reddy et al., 2019

Source	Licence	Citation
DISPLACE-M challenge paper	Data under signed terms	IISc LEAP Lab, 2026
DISPLACE 2024 paper	Not stated	2024
IndicVoices	CC-BY-4.0	AI4Bharat
ASR Under the Stethoscope	–	arXiv 2512.10967
Partner's research repo	–	L9G9N0/voice_scribe